Math 358: Advanced Calculus Summary
Midterm Date: 26 February 2026 1.00 - 2.30
Key Results, Theorems, Definitions, etc.

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Abstract

This document contains a summary of all the key definitions, results, and theorems from class before the midterm. Point-set topology in Euclidean space; continuity and differentiability of functions in several variables. Implicit and inverse function theorems, Lagrange multipliers, etc...

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1. Continuity and Differentiability

1.1. Differentiability

Definition 1 (Differentiation)**.** $f : \Omega \to \mathbb{R}^m$ where $\Omega \subset \mathbb{R}^n$ is a domain[1], $f := (f_1, \dots, f_m)$, $f_i : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is **differentiable** at $x_0 \in \Omega$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0$$

Equivalently, $\forall \epsilon > 0\ \exists \delta > 0$ such that for any $x \in \mathbb{R}$ with $\|x - x_0\| < \delta$ [2] we have

$$\|f(x) - f(x_0) - L(x - x_0)\| \leq \epsilon \|x - x_0\|$$

Theorem 1. If $f : \Omega \to \mathbb{R}^m$ is differentiable at $x_0 \in \Omega$, then the linear map L is unique.

Definition 2. If f differentiable at $x_0 \in \mathbb{R}$, with $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ we write the differential map of f at $x_0 \in \Omega$ as $L = Df(x_0)$

Proposition 1. If f is differentiable at $x_0 \in \Omega$, then f is continuous at $x_0 \in \Omega$. In fact, it is Lipschitz continuous at x_0, i.e. $\exists k > 0$ and $\delta > 0$ such that if $\|x - x_0\| < \delta$, then $\|f(x) - f(x_0)\| \leq k\|x - x_0\|$

Proposition 2. If $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ differentiable at $x_0 \in \Omega$, and $f = (f_1, \dots, f_m)$, $f_i : \Omega \to \mathbb{R}$, then f differentiable at $x_0 \in \Omega$ iff $f_j : \Omega \to \mathbb{R}$ is differentiable at $x_0 \in \Omega$ for all $j = 1, \dots, m$.

Definition 3 (Partial Derivatives)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$, $f(x_1, \dots, x_m) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$. Then

$$\frac{\partial f_j}{\partial x_i}(x_0) := \lim_{h \to 0} \frac{[f_j(x_1, \dots, x_i + h, \dots, x_n) - f_j(x_1, \dots, x_i, \dots, x_n)]}{h}$$

if the limit exists. We call $\frac{\partial f_j}{\partial x_i}(x_0)$ the **partial derivative** of f_j with respect to x_i at x_0.

[1] Open, Connected

[2] Here, $\|\cdot\|$ is the standard Euclidean norm on $\mathbb{R}^m$ and $\mathbb{R}^n$.

Proposition 3. Let $f : \Omega \subset \mathbb{R}^m \to \mathbb{R}^m$, $f = (f_1, \dots, f_m)$, $f_j : \Omega \to \mathbb{R}$, be differentiable at $x_0 \in \Omega$. Then all partial derivatives $\frac{\partial f_j}{\partial x_i}(x_0)$ exist and

$$L = Df(x_0) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \frac{\partial f_1}{\partial x_2}(x_0) & \dots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \frac{\partial f_2}{\partial x_2}(x_0) & \dots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \frac{\partial f_m}{\partial x_2}(x_0) & \dots & \frac{\partial f_m}{\partial x_n}(x_0) \end{bmatrix}$$

called the Jacobian of f at x_0[3].

Remark. f differentiable at $x_0 \in \Omega \Rightarrow \frac{\partial f_j}{\partial x_i}(x_0)$ exists for all i, j. However, the converse need not be true, for example the classic example of $f : \mathbb{R}^2 \to \mathbb{R}$,

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Both partial derivatives of f at $(0, 0)$ exist and are equal to 0, but f is not differentiable at $(0, 0)$.

Steps to solve: Find candidate for $Df(0, 0)$ using the partial derivatives, then show that the limit in the definition of differentiability does not go to 0 for this candidate.

Theorem 2. Let $f = (f_1, \dots, f_m) : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$. Suppose that all partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist and are continuous at $x_0 \in \Omega$. Then f is differentiable at x_0.

Definition 4. Suppose $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ has continuous partial derivatives $\frac{\partial f}{\partial x_i}$, $i = 1, \dots, n$ at all points $x \in \Omega$. Then f is continuously differentiable on Ω, and we write $f \in C^1(\Omega)$.

Remark. Continuity of partial derivatives is a sufficient condition for differentiability, but it is not necessary. For example, the function $f : \mathbb{R}^2 \to \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Then $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$, however, along the line $x = y^2, x = t, y = t^2$, $\frac{\partial f}{\partial x}(t^2, t) = \frac{1}{2} \neq 0$. But, f is differentiable at $(0, 0)$ with $Df(0, 0) = 0$.

Proposition 4 (Basic Properties of Differentiation)**.**

(1) $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ both differentiable at $x_0 \in \Omega$, then so is $F = f + g$. Moreover, $DF(x_0) = Df(x_0) + Dg(x_0)$.

(2) $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ both differentiable at $x_0 \in \Omega$, then so is $F = f \cdot g : \Omega \to \mathbb{R}$. Moreover, $DF(x_0) = f(x_0) \cdot Dg(x_0) + g(x_0) \cdot Df(x_0)$.

(3) $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ both differentiable at $x_0 \in \Omega$, and $g(x_0) \neq 0$, then so is $F = \frac{f}{g} : \Omega \to \mathbb{R}$. Moreover,
$$DF(x_0) = \frac{g(x_0) \cdot Df(x_0) - f(x_0) \cdot Dg(x_0)}{g(x_0)^2}$$

(4) $f : \Omega \subset \mathbb{R}^n \to \tilde{\Omega} \subset \mathbb{R}^m$, $g : \tilde{\Omega} \subset \mathbb{R}^m \to \mathbb{R}^k$ with f differentiable at $x_0 \in \Omega$, and g differentiable at $y_0 = f(x_0) \in \tilde{\Omega}$. Then $H = g \circ f$ is differentiable at $x_0 \in \Omega$ and
$$DH(x_0) = Dg(y_0) \cdot Df(x_0)$$

[3] also called the derivative of f at $x_0 \in \Omega$

Theorem 3 (Chain Rule)**.** Let $f : \Omega \subset \mathbb{R}^n \to \tilde{\Omega} \subset \mathbb{R}^m$, $\tilde{\Omega} = f(\Omega)$, $g : \tilde{\Omega} \subset \mathbb{R}^m \to R^k$. If f is differentiable at $x_0 \in \Omega$ and g is differentiable at $y_0 = f(x_0) \in \tilde{\Omega}$, then $H = g \circ f : \Omega \subset \mathbb{R}^n \to R^k$ is differentiable at $x_0 \in \Omega$, and

$$DH(x_0) = Dg(y_0) \cdot Df(x_0)$$

Example 1.1. Given f differentiable in R^2 and $g(r, \theta) = (r\cos\theta, r\sin\theta)$, with $(r, \theta) \in (0, \infty) \times [0, 2\pi]$. Set $F(r, \theta) = f(g(r, \theta))$, compute $\frac{\partial F}{\partial r}(r, \theta)$ and $\frac{\partial F}{\partial \theta}(r, \theta)$ in terms of the partial derivatives of f, using the chain rule.

To solve, the multivariable chain rule gives us

$$\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r} = \frac{\partial f}{\partial x}\cos\theta + \frac{\partial f}{\partial y}\sin\theta$$

and

$$\frac{\partial F}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta} = -r\frac{\partial f}{\partial x}\sin\theta + r\frac{\partial f}{\partial y}\cos\theta$$

1.2. Tangent Planes and Directional Derivatives

Definition 5 (Gradient)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable on Ω. Then $Df(x) = (\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_n}(x)) = \nabla f(x)$ is called the **gradient** of f at $x \in \Omega$.

Definition 6 (Tangent Planes)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable on Ω. $S := \operatorname{graph} f = \{(x, z) \in \Omega \times \mathbb{R} : z = f(x)\}$. Then for $x_0 \in \Omega$, $T_{x_0}S = \{(x, z) \in \mathbb{R}^n \times \mathbb{R} : z = f(x_0) + Df(x_0) \cdot (x - x_0)\}$

Definition 7 (Directional Derivative)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable on Ω. Then for $x_0 \in \Omega$ and $v \in \mathbb{R}^n$, the directional derivative of f at x_0 in the direction of v is defined as

$$D_v f(x_0) = \langle \nabla f(x_0), v \rangle$$

Proposition 5. Suppose $\nabla f(x) \neq 0$. Then $\nabla f(x)$ points in the direction of the steepest increase of f. [4]

1.3. Clairaut's Theorem

Theorem 4 (MVT in $\mathbb{R}^n$)**.** Let $B \subset \mathbb{R}^n$ be a ball, $B = \{x \in \mathbb{R}^n : \|x - x_0\| < p \text{ for some } x_0\}$, and $f : B \to \mathbb{R}$ differentiable on B. Then, for any $x, y \in B$ there exists $z \in B$ such that

$$f(y) - f(x) = Df(z) \cdot (y - x)$$

As a consequence, $\|f(y) - f(x)\| \leq \|Df(z)\| \cdot \|y - x\|$.

Definition 8. We say f twice differentiable at $x \in \Omega$ if Df exists locally near $x \in \Omega$ and $Df : \Omega \subset \mathbb{R}^n \to \mathbb{R}^{n \times m}$ is differentiable at $x \in \Omega$. Then $D^2 f = D(Df)$ at $x \in \Omega$. Similarly, for all $k \geq 2$, $D^k f = D(D^{k-1} f)$.

Definition 9. Given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, we say that $f \in C^k(\Omega)$ if all the partial derivatives up to order k are continuous on Ω.

Definition 10. (Hessian) Given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, the **Hessian** matrix of f at $x \in \Omega$ is

$$Hf(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(x) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(x) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(x) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(x) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(x) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(x) & \frac{\partial^2 f}{\partial x_n \partial x_2}(x) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(x) \end{bmatrix}$$

[4] Proof follows from the Cauchy-Schwarz inequality, $< a, b >= \|a\| \cdot \|b\| \cos\theta$

For $f(x,y)$, $Df = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ and

$$D^2 f = Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x \partial x} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y \partial y} \end{bmatrix}$$

Theorem 5 (Clairaut's Theorem)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at $x \in \Omega$. Then,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$$

for all $i, j = 1, \ldots, n$. Moreover, if $f \in C^2(\Omega)$, then the above holds for all $x \in \Omega$.

Corollary 1. If $\frac{\partial^2 f}{\partial x_i \partial x_j}(x)$ are all continuous at $x \in \Omega$ for all $i, j = 1, \ldots, n$, then $\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$ for all $i, j = 1, \ldots, n$.

1.4. Inverse Function Theorem and Implicit Function Theorem

Theorem 6 (Banach Fixed Point)**.** Let (X, d) be a complete metric space, and $f : X \to X$ be a contraction mapping with $d(f(x), f(y)) \leq \alpha d(x, y)$ for some $0 < \alpha < 1$. Then $\exists$ unique $x_0 \in X$ such that $f(x_0) = x_0$.

Definition 11. Let $M_n := \{n \times n \text{ matrices}\} \cong \mathbb{R}^{n^2}$ and $\|A\| := \sqrt{\sum_{i,j} a_{ij}^2}$ for $A = (a_{ij}) \in M_n$.

Let $GL_n := \{A \in M_n : \det A \neq 0\} = \det^{-1}(\mathbb{R} \setminus \{0\})$ Thus since $\det : M_n \to \mathbb{R}$ is continuous, GL_n is an open subset of M_n.

Theorem 7 (Inverse Function)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ be C^1. Let $x_0 \in \Omega$ and assume $Df(x_0) \in GL_n$. Then there exists neighbourhoods U and V of x_0 and $f(x_0)$ respectively such that $f(U) = V$ and $f|_U$ has a C^1 inverse map $f^{-1} : V \to U$. Moreover, for any $y \in V$, and $x = f^{-1}(y)$, we have $Df^{-1}(y) = [Df(x)]^{-1}$.

Remark. If $f \in C^k$, $k \geq 1$, then the inverse map f^{-1} is also in C^k.

Example 1.2. IFT is local. For example, $f(x,y) = (e^y \cos x, e^y \sin x)$. Then $Df(x,y) = \begin{bmatrix} -e^y \sin x & e^y \cos x \\ e^y \cos x & e^y \sin x \end{bmatrix}$, and $\det Df(x,y) = -(\cos^2 x + \sin^2 x)e^{2y} = -e^{2y} \neq 0$ for all $(x,y) \in \mathbb{R}^2$. However, $f(x + 2k\pi, y) = f(x,y)$ for all $k \in \mathbb{Z}$, so f is not injective globally on (x,y).

Theorem 8 (Implicit Function Theorem)**.** $F : \Omega \subset \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ C^k map, $X = (x,y) \in R^n \times \mathbb{R}^m$ and $X_0 = (x_0, y_0)$ with $F(X_0) = 0$. Writing $F = (F_1, \ldots, F_m)$, assume that

$$D_y F(X_0) = \begin{bmatrix} \frac{\partial F_1}{\partial y_1} & \frac{\partial F_1}{\partial y_2} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \frac{\partial F_2}{\partial y_1} & \frac{\partial F_2}{\partial y_2} & \cdots & \frac{\partial F_2}{\partial y_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \frac{\partial F_m}{\partial y_2} & \cdots & \frac{\partial F_m}{\partial y_m} \end{bmatrix} (X_0) \in GL_m$$

Then there exists neighbourhoods U and V of $x_0 \in \mathbb{R}^n$ and $y_0 \in \mathbb{R}^m$ and a unique C^k map $f : U \to V$ such that $F(x, f(x)) = 0$, for all $x \in U$.

Corollary 2. Let $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be $C^k(\Omega)$. Let $x = (x', y) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and suppose $(x_0', y_0) \in \Omega$ with $\frac{\partial f}{\partial y}(x_0', y_0) \neq 0$. Then there exists neighbourhoods U, V of $x_0' \in \mathbb{R}^{n-1}$ and $y_0 \in \mathbb{R}$ and a unique function $f : U \to V$, $f \in C^k(U)$ such that $\{F(x', y) = 0\} = \{y = f(x')\}\, \forall x' \in U$.

Theorem 9 (Morse Lemma)**.** $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ a C^k function $k \geq 3$. Let $0 \in \Omega$ be a critical point, i.e. $\nabla f(0) = 0$, and assume $f(0) = 0, \nabla f(0) = 0, Hf(0) \in GL_n$. Then there exists open sets U and V of $0 \in U \cap V$ and $g \in C^{k-2}(U)$, $g : U \to V$ with $g^{-1} : U \to V$, $g^{-1} \in C^2(V)$ such that $f(g(y)) = y_{\ell+1}^2 + \cdots + y_n^2 - (y_1^2 + y_2^2 + \ldots y_\ell^2)$. Here $\ell \in \mathbb{Z}$, $\ell \in [0, n]$.

1.5. Taylor's Theorem

Theorem 10 (Taylor's Theorem in $\mathbb{R}^n$). Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $f \in C^{k+1}(\Omega)$. Then, for any $x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^{k} \frac{1}{j!} \left[\sum_{i_1=1}^{n} \cdots \sum_{i_j=1}^{n} \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x^0)(x_{i_1} - x^0_{i_1}) \dots (x_{i_j} - x^0_{i_j}) \right] + R_k(f)(x)$$

where $\frac{|R_k(f)(x)|}{\|x-x^0\|^k} \leq M_k \|x - x^0\|$ as $x \to x^0$.

Remark. The k-th order Taylor polynomial of f at x^0 is the sum of the first two terms in the above expansion, i.e.

$$P_k(f)(x) = f(x^0) + \sum_{j=1}^{k} \frac{1}{j!} \left[\sum_{i_1=1}^{n} \cdots \sum_{i_j=1}^{n} \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x^0)(x_{i_1} - x^0_{i_1}) \dots (x_{i_j} - x^0_{i_j}) \right]$$

This is a good approximation to $f(x)$ near x^0 provided $\lim_{x \to x^0} \frac{|R_k(f)(x)|}{\|x-x^0\|^k} = 0$.

Theorem 11. Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be $C^{k-1}(\Omega)$ and assume $D^{k-1}f$ is differentiable at $x^0 \in \Omega$. Then the k-th order Taylor expansion in the previous theorem still holds, but with the only assumption of the remainder term being $\lim_{x \to x^0} \frac{|R_k(f)(x)|}{\|x-x^0\|^k} = 0$.

Example 1.3. Let $f(x, y) = e^x + sin(xy)$. Compute the second order Taylor expansion of f around $(0, 0)$.

To solve this problem, we first recognize that $f \in C^\infty(\mathbb{R}^2)$, so we can apply Taylor's theorem. We compute the first and second order partial derivatives of f at $(0, 0)$:

$$f_x = \frac{\partial f}{\partial x} = e^x + y\cos(xy), \quad f_y = \frac{\partial f}{\partial y} = x\cos(xy)$$

Therefore $f_x(0, 0) = 1$ and $f_y(0, 0) = 0$. Next, we compute the second order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = e^x - y^2 \sin(xy), \quad f_{yy} = \frac{\partial^2 f}{\partial y^2} = -x^2 \sin(xy), \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \cos(xy) - xy\sin(xy)$$

Therefore, $f_{xx}(0, 0) = 1$, $f_{yy}(0, 0) = 0$, and $f_{xy}(0, 0) = 1$. Thus, the second order Taylor expansion of f around $(0, 0)$ is

$$f(x, y) = f(0, 0) + f_x(0, 0)x + f_y(0, 0)y + \frac{1}{2!}\left[f_{xx}(0, 0)x^2 + 2f_{xy}(0, 0)xy + f_{yy}(0, 0)y^2\right] + R_2(f)(x, y)$$

which simplifies to

$$f(x, y) = 1 + x + \frac{x^2}{2} + xy + R_2(f)(x, y)$$

and the remainder term satisfies $\frac{|R_2(f)(x,y)|}{x^2+y^2} \leq M\sqrt{x^2 + y^2}$ as $(x, y) \to (0, 0)$.

1.6. Lagrange Multipliers

Theorem 12 (Lagrange Multiplier). Let $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be C^1 functions in $\mathbb{R}$. Let $\Sigma = \{x \in \Omega : g(x) = 0\}$ and assume that $Dg(x_0) \neq 0$ for all $x_0 \in \Sigma$. Then if $f|_\Sigma$ has a max or min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda g(x_0)$$

Theorem 13 (Max/Min). Let $\tilde{\Omega} \supset \Omega$ be a slightly larger domain and assume $f, g \in C^1(\Omega)$, where $\partial\Omega = \left\{x \in \tilde{\Omega} : g(x) = 0\right\}$, $\nabla g(x) \neq 0 \forall x \in \partial\Omega$. Then,

(1) If $x_0 \in \partial\Omega$ is a local max/min of $f|_{\partial\Omega}$, then $\nabla f(x_0) = \lambda \nabla g(x_0)$ for some $\lambda \in \mathbb{R}$.

(2) If $x_0 \in \mathrm{Int}(\Omega)$ is a local max/min of f in $\mathrm{Int}(\Omega)$, then $\nabla f(x_0) = 0$.

(3) Global max/min of $f|_{\tilde{\Omega}}$ is largest/smallest value of f at all points in 1 and 2.

Theorem 14 (Multiple Constraints)**.** $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $g_i : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $i = 1, \dots, k$ are C^1 functions. Let $\Gamma_0 = \{x \in \Omega : g_i(x) = 0, i = 1, \dots, k\}$ and assume that $\{\nabla g_1(x), \dots, \nabla g_k(x)\}$ are linearly independent for all $x \in \Gamma_0$. Then if f_{Γ_0} has a max/min at $x_0 \in \Gamma_0$, then there exists $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ such that

$$\nabla f(x_0) = \sum_{i=1}^{k} \lambda_i \nabla g_i(x_0)$$

Example 1.4. Find the absolute max and min of $f(x_1, \dots, x_n) = x_1 + x_2$ on $B_1(0) = \left\{|x|^2 \le 1\right\} = \left\{x_1^2 + \cdots + x_n^2 \le 1\right\}$, $\partial B_1(0) = \left\{x_1^2 + \cdots + x_n^2 = 1\right\}$.

To solve, first observe $f \in C^k(\mathbb{R}^n)$ for all $k \ge 1$. Thus we will apply the max/min theorem with $g(x) = |x|^2 - 1 = x_1^2 + \cdots + x_n^2 - 1$. Then, since $\nabla f(x) = (1, 1, 0, \dots, 0) \neq 0$, there are no critical points in $\mathrm{Int}(B_1(0)) = \left\{|x|^2 < 1\right\}$. So the global max/min must occur on $\partial B_1(0)$.

By Lagrange multiplier theorem, at max/min $x_0 \in \partial B_1(0)$, there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0) \Leftrightarrow (1, 1, 0, \dots, 0) = \lambda(2x_1^0, 2x_2^0, 2x_3^0, \dots, 2x_n^0) \tag{1}$$

$$\Leftrightarrow x_1^0 = x_2^0 = \frac{1}{2\lambda}, x_3^0 = \cdots = x_n^0 = 0 \tag{2}$$

Also,

$$(x_1^0)^2 + (x_2^0)^2 = 1 \Leftrightarrow \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 1 \Leftrightarrow \lambda = \pm\frac{1}{\sqrt{2}}$$

Thus, the candidates for max/min are $x_0 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0\right)$ and $x_0 = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots, 0\right)$. Evaluating f at these points, we get

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0\right) = \sqrt{2}, \quad f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots, 0\right) = -\sqrt{2}$$

Thus, the absolute max of f on $B_1(0)$ is $\sqrt{2}$ and the absolute min is $-\sqrt{2}$.

1.7. Doneso